CFM301 Project 3: Reproducing the Equity Return Predictability Table

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1. Economic question and audience

Question: does information available at month-end predict future market excess returns, and is that predictability still present after 2011? We reproduce the predictability table of Chapter 8 (slides 24-25 and the surrounding blocks): correlations and Newey-West t-statistics between each predictor's level today and future equity excess returns at 1-quarter, 1-year, 2-year and 5-year horizons, extended to the longest sample the submitted data allow (1926-01 through 2026-10). The audience is pension-fund and other long-horizon asset-allocation managers deciding whether a valuation, business-cycle or risk signal justifies adjusting equity exposure. The market series is CRSP value-weighted total return (vwretd), not the ex-dividend return. Future returns overlap at longer horizons, so t-statistics use Newey-West/HAC covariance with horizon - 1 lags.

2. Data, construction and cleaning

All results come from the submitted data used in docs/predictors.ipynb. Valuation ratios (annual dividend yield and earnings yield) come from market_data_full_history_2025.csv; market returns from Market_Dividends_Returns.csv (CRSP vwretd). Macro and financial predictors come from FRED files: GS10 and TB3MS (term spread = 10y - 3m), BAA and AAA (credit spread = BAA - AAA), CPIAUCSL inflation (past year), INDPRO growth (past year), UNRATE, GDPC1 growth (past year, quarterly series carried forward), DCOILWTICO oil-price change (past year), VIXCLS, and the 3-month T-bill rate. The consumption-wealth ratio is the Lettau-Ludvigson cay series as supplied (cay_current.csv, c - 0.218a - 0.801y + 0.441), with the quarterly series carried forward from the last observed quarter-end. All sources are converted to month-end dates and duplicate dates are removed before joining (0 duplicates removed). Excess returns are geometric: (1 + total return) / (1 + monthly T-bill rate) - 1, where the monthly rate is (1 + TB3MS/100)^(1/12) - 1. Past 1-year return is the trailing 12-month compounded return; past volatility (last quarter) is the standard deviation of monthly returns over the last 3 months, annualized by sqrt(12) (correlation is scale-free, so annualizing changes nothing). Predictors are winsorized at the 1st and 99th percentiles within each usable estimation sample; returns are not winsorized. Future returns are overlapping compounded returns, (1 + r) rolled over h months, shifted forward by h months, minus 1. Each cell is an OLS regression of the future excess return on the predictor level, with a HAC t-statistic at horizon - 1 lags; |t| >= 2 is treated as significant.

Missing values before estimation (months with no observation): oil 860, VIX 815, term spread 328, cay 314, unemployment 266, inflation 266, GDP 264, risk-free rate 97, past 1-year return 33, past volatility 24, industrial production 13, dividend yield 10, earnings yield 10, credit spread 1. The gaps reflect series start dates (oil from 1986, VIX from 1990, quarterly series carried forward) and are handled by pairwise deletion, so each cell uses only months where both predictor and future return exist. Estimation sample sizes and winsorized counts:

predictor	1Q N/clip	1Y N/clip	2Y N/clip	5Y N/clip
Dividend yield	1090/22	1081/22	1069/22	1033/22
Earnings yield	1090/22	1081/22	1069/22	1033/22
Term spread (10y - 3m)	858/18	849/18	837/18	801/16
Credit spread (BAA - AAA)	1090/22	1081/22	1069/22	1033/22
Unemployment rate	921/18	912/18	900/18	864/16
GDP growth (past year)	921/18	912/18	900/18	864/16
Industrial production (past ye	1090/22	1081/22	1069/22	1033/22
Past 1-year return	1090/22	1081/22	1069/22	1033/22
Past volatility (last quarter)	1090/22	1081/22	1069/22	1033/22
Inflation (past year)	921/20	912/20	900/18	864/18
Oil price (past-year change)	329/ 8	322/ 8	311/ 8	279/ 6
Risk-free rate	1089/15	1080/15	1068/15	1032/15
VIX	372/ 8	364/ 8	353/ 8	319/ 8
Consumption-wealth ratio	871/18	862/18	850/18	814/18

3. Main table: future excess returns (correlation, HAC t)

predictor	1Q	1Y	2Y	5Y
Dividend yield	+0.11 (+2.40)*	+0.25 (+3.06)*	+0.34 (+3.28)*	+0.53 (+3.87)*
Earnings yield	+0.09 (+2.03)*	+0.21 (+2.69)*	+0.27 (+2.64)*	+0.42 (+2.31)*
Term spread (10y - 3m)	+0.08 (+1.63)	+0.17 (+2.06)*	+0.21 (+2.45)*	+0.38 (+3.15)*
Credit spread (BAA - AAA)	+0.07 (+1.08)	+0.14 (+1.62)	+0.13 (+1.05)	+0.09 (+0.65)
Unemployment rate	+0.15 (+3.28)*	+0.22 (+2.78)*	+0.14 (+1.34)	+0.22 (+1.26)
GDP growth (past year)	-0.12 (-2.20)*	-0.25 (-2.86)*	-0.19 (-2.08)*	-0.19 (-1.49)
Industrial production (past year)	-0.10 (-1.73)	-0.16 (-1.72)	-0.05 (-0.47)	-0.00 (-0.04)
Past 1-year return	-0.04 (-0.66)	-0.16 (-2.23)*	-0.24 (-3.88)*	-0.20 (-2.35)*
Past volatility (last quarter)	+0.12 (+2.48)*	+0.06 (+1.41)	+0.06 (+1.01)	+0.05 (+0.65)
Inflation (past year)	-0.14 (-2.52)*	-0.21 (-2.24)*	-0.21 (-2.56)*	-0.17 (-2.03)*
Oil price (past-year change)	-0.21 (-2.41)*	-0.34 (-2.52)*	-0.34 (-3.51)*	-0.23 (-2.38)*
Risk-free rate	-0.10 (-2.10)*	-0.16 (-1.82)	-0.21 (-2.18)*	-0.31 (-1.91)
VIX	+0.13 (+1.38)	+0.18 (+1.47)	+0.17 (+1.16)	+0.13 (+0.40)
Consumption-wealth ratio	+0.09 (+1.72)	+0.19 (+2.01)*	+0.29 (+2.44)*	+0.33 (+2.16)*

* marks |t| >= 2. Valuation ratios are positive at every horizon and grow with horizon: dividend yield 0.11/0.25/0.34/0.53, earnings yield 0.09/0.21/0.27/0.42. Business-cycle signs are mostly as the slides predict (term spread and unemployment positive; GDP growth, industrial production, past return, inflation, oil and the risk-free rate negative), with oil and inflation the strongest negatives. Volatility on its own is weak (VIX 0.13-0.18, past volatility 0.05-0.12, significant only at 1 quarter for past volatility). cay is positive and significant from 1 year on (0.19/0.29/0.33, t = 2.01/2.44/2.16) but not at 1 quarter (0.09, t = 1.72).

4. Comparison with the slides

Valuation ratios (sign agreement 8/8)

Dividend yield	1Q	slide +0.12 (+1.12)	vs	notebook +0.11 (+2.40)
Dividend yield	1Y	slide +0.23 (+1.25)	vs	notebook +0.25 (+3.06)
Dividend yield	2Y	slide +0.33 (+1.48)	vs	notebook +0.34 (+3.28)
Dividend yield	5Y	slide +0.49 (+1.95)	vs	notebook +0.53 (+3.87)
Earnings yield	1Q	slide +0.18 (+2.23)	vs	notebook +0.09 (+2.03)
Earnings yield	1Y	slide +0.33 (+2.55)	vs	notebook +0.21 (+2.69)
Earnings yield	2Y	slide +0.41 (+2.79)	vs	notebook +0.27 (+2.64)
Earnings yield	5Y	slide +0.48 (+2.31)	vs	notebook +0.42 (+2.31)

Business cycle and markets (sign agreement 33/36)

Term spread (10y-3m)	1Q	slide +0.11 (+1.60)	vs	notebook +0.08 (+1.63)
Term spread (10y-3m)	1Y	slide +0.23 (+1.84)	vs	notebook +0.17 (+2.06)
Term spread (10y-3m)	2Y	slide +0.27 (+1.66)	vs	notebook +0.21 (+2.45)
Term spread (10y-3m)	5Y	slide +0.35 (+1.61)	vs	notebook +0.38 (+3.15)
Credit spread (BAA-AAA)	1Q	slide +0.02 (+0.15)	vs	notebook +0.07 (+1.08)
Credit spread (BAA-AAA)	1Y	slide +0.08 (+0.34)	vs	notebook +0.14 (+1.62)
Credit spread (BAA-AAA)	2Y	slide +0.12 (+0.44)	vs	notebook +0.13 (+1.05)
Credit spread (BAA-AAA)	5Y	slide +0.20 (+0.67)	vs	notebook +0.09 (+0.65)
Unemployment rate	1Q	slide +0.12 (+1.86)	vs	notebook +0.15 (+3.28)
Unemployment rate	1Y	slide +0.22 (+1.85)	vs	notebook +0.22 (+2.78)
Unemployment rate	2Y	slide +0.19 (+1.08)	vs	notebook +0.14 (+1.34)
Unemployment rate	5Y	slide +0.22 (+0.75)	vs	notebook +0.22 (+1.26)
GDP growth (past year)	1Q	slide +0.01 (+0.17)	vs	notebook -0.12 (-2.20)
GDP growth (past year)	1Y	slide -0.13 (-1.37)	vs	notebook -0.25 (-2.86)
GDP growth (past year)	2Y	slide -0.11 (-1.25)	vs	notebook -0.19 (-2.08)
GDP growth (past year)	5Y	slide -0.08 (-0.84)	vs	notebook -0.19 (-1.49)
Industrial production (p	1Q	slide -0.01 (-0.06)	vs	notebook -0.10 (-1.73)
Industrial production (p	1Y	slide -0.04 (-0.25)	vs	notebook -0.16 (-1.72)
Industrial production (p	2Y	slide +0.02 (+0.08)	vs	notebook -0.05 (-0.47)
Industrial production (p	5Y	slide -0.04 (-0.20)	vs	notebook -0.00 (-0.04)
Past 1-year return	1Q	slide -0.01 (-0.12)	vs	notebook -0.04 (-0.66)
Past 1-year return	1Y	slide -0.05 (-0.26)	vs	notebook -0.16 (-2.23)

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Past 1-year return       2Y  slide -0.13 (-0.55)  vs  notebook -0.24 (-3.88)
Past 1-year return       5Y  slide -0.26 (-1.39)  vs  notebook -0.20 (-2.35)
Inflation (past year)    1Q  slide -0.07 (-1.07)  vs  notebook -0.14 (-2.52)
Inflation (past year)    1Y  slide -0.12 (-0.94)  vs  notebook -0.21 (-2.24)
Inflation (past year)    2Y  slide -0.10 (-0.58)  vs  notebook -0.21 (-2.56)
Inflation (past year)    5Y  slide -0.03 (-0.09)  vs  notebook -0.17 (-2.03)
Oil price (past-year cha 1Q  slide -0.10 (-1.20)  vs  notebook -0.21 (-2.41)
Oil price (past-year cha 1Y  slide -0.18 (-1.20)  vs  notebook -0.34 (-2.52)
Oil price (past-year cha 2Y  slide -0.16 (-1.08)  vs  notebook -0.34 (-3.51)
Oil price (past-year cha 5Y  slide -0.11 (-0.78)  vs  notebook -0.23 (-2.38)
Risk-free rate           1Q  slide -0.07 (-1.23)  vs  notebook -0.10 (-2.10)
Risk-free rate           1Y  slide -0.12 (-1.17)  vs  notebook -0.16 (-1.82)
Risk-free rate           2Y  slide -0.15 (-1.05)  vs  notebook -0.21 (-2.18)
Risk-free rate           5Y  slide -0.22 (-0.97)  vs  notebook -0.31 (-1.91)
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Risk and risk aversion (sign agreement 11/12)

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VIX                      1Q  slide +0.12 (+0.96)  vs  notebook +0.13 (+1.38)
VIX                      1Y  slide +0.10 (+0.63)  vs  notebook +0.18 (+1.47)
VIX                      2Y  slide +0.13 (+0.75)  vs  notebook +0.17 (+1.16)
VIX                      5Y  slide -0.16 (-0.47)  vs  notebook +0.13 (+0.40)
Past volatility (last qu 1Q  slide +0.05 (+0.55)  vs  notebook +0.12 (+2.48)
Past volatility (last qu 1Y  slide +0.09 (+0.65)  vs  notebook +0.06 (+1.41)
Past volatility (last qu 2Y  slide +0.13 (+0.78)  vs  notebook +0.06 (+1.01)
Past volatility (last qu 5Y  slide +0.06 (+0.17)  vs  notebook +0.05 (+0.65)
Consumption-wealth ratio 1Q  slide +0.16 (+2.92)  vs  notebook +0.09 (+1.72)
Consumption-wealth ratio 1Y  slide +0.30 (+2.62)  vs  notebook +0.19 (+2.01)
Consumption-wealth ratio 2Y  slide +0.42 (+2.23)  vs  notebook +0.29 (+2.44)
Consumption-wealth ratio 5Y  slide +0.53 (+1.34)  vs  notebook +0.33 (+2.16)
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Overall sign agreement is 53 of 56 cells. Magnitudes are close for dividend yield (5-year 0.53 vs slide 0.49), term spread (0.38 vs 0.35), unemployment (0.22 vs 0.22) and credit spread at short horizons. Three gaps stand out. First, earnings yield is about 0.1 lower than the slides at every horizon (0.09/0.21/0.27/0.42 vs 0.18/0.33/0.41/0.48) -- and this notebook uses a different earnings source than the earlier Shiller-based pipeline, which gave nearly the same numbers (0.09/0.20/0.26/0.38), so the gap is not an artefact of one earnings series. Second, oil is about twice the slides' magnitude on monthly data (-0.34/-0.34 vs -0.18/-0.16 at 1Y/2Y); quarterly sampling closes most of that gap in the earlier pipeline, suggesting the slides use quarterly observations for this block. Third, cay is weaker here than in the slides (5-year 0.33 vs 0.53), the reverse of the earlier DOLS-reconstructed cay (0.67): the supplied cay series uses fixed Lettau-Ludvigson coefficients rather than re-estimated ones, and that choice matters for the level of cay predictability. VIX at 5 years flips sign (+0.13 vs slide -0.16) but is insignificant in both. The t-statistics here are generally larger than the slides' (for example dividend yield 5-year 3.87 vs 1.95), consistent with the slides using a more conservative standard-error construction; Section 8 shows how much the overlap correction itself matters.

5. Interpretation: the dividend discount model

From class, P = D / (E(r) - g), so rearranging gives D/P = E(r) - g. Holding expected dividend growth g approximately constant, a high dividend yield means investors require a higher expected return. The same logic applies to the earnings yield: a high E/P means the price is low relative to the earnings the market generates, consistent with a higher required and expected future return. The extended table supports this prediction: dividend yield and earnings yield have positive correlations with future excess returns, and the correlations generally increase from one quarter to five years. Their post-2011 observations are included in the extended sample through the latest available market-return date (2026-10). Long-horizon predictability is stronger because valuation ratios move slowly while returns compound: a small difference in expected monthly returns accumulates over 12, 24 or 60 months, whereas one-quarter returns contain much more noise. The longer-horizon estimates should still be interpreted cautiously because overlapping observations reduce the effective sample size; the table therefore reports HAC t-statistics.

6. Five-bucket valuation analysis

Following slides 26-28, observations are sorted into five equal-count buckets by dividend yield at time t. Q1 is the lowest-yield (most expensive) group and Q5 the highest-yield (cheapest) group. The table shows average compounded 1-year returns and average annualized 5-year returns with +/- 2 standard-deviation bands. These bands are dispersion bands, not confidence intervals.

bucket	avg yield	1Y mean (SD)	5Y ann. mean (SD)	n
Q1 lowest	1.70%	3.91% (13.03%)	3.58% (4.86%)	207
Q2	2.61%	4.51% (16.28%)	7.10% (6.83%)	206
Q3	3.38%	7.39% (15.58%)	4.70% (5.46%)	207
Q4	4.34%	9.15% (19.93%)	6.73% (4.84%)	206
Q5 highest	6.00%	16.79% (17.39%)	13.27% (5.45%)	207

The cheapest bucket beats the most expensive bucket by a wide margin: 1-year 16.79% vs 3.91%, 5-year annualized 13.27% vs 3.58%. But the middle buckets are not monotonic at 5 years (Q2 7.10% above Q3 4.70% and Q4 6.73%), and the +/- 2 SD bands overlap heavily across buckets (for example Q5 1-year 16.79% with SD 17.39%): even where mean returns differ, the distribution of outcomes within each bucket is wide. That is why a statistically visible correlation is still too weak to trade on aggressively -- the slides' verdict for the business-cycle block, and it applies to valuation timing too.

7. Returns versus volatility

The slides emphasize that volatility is more predictable than returns. VIX at time t against annualized realized volatility over the following horizon gives:

horizon	corr (HAC t)	n
1Q	+0.29 (+4.90)	372
1Y	+0.38 (+4.20)	364
2Y	+0.30 (+3.12)	353
5Y	-0.16 (-1.04)	319

VIX predicts realized volatility up to 2 years (correlations 0.29/0.38/0.30, all significant) but not at 5 years (-0.16, t = -1.04). A GARCH(1,1) model on monthly total returns tells the same story from the other side. Estimated by Gaussian quasi-maximum likelihood (returns x 100), h_t = omega + alpha r2_{t-1} + beta h_{t-1} gives omega = 0.7153, alpha = 0.1229 (ARCH response to new shocks) and beta = 0.8538 (GARCH persistence), persistence alpha + beta = 0.9767. Annualized GARCH volatility averages 17.09% (SD 7.27%) against next-quarter realized volatility averaging 13.92% (SD 10.93%), correlation 0.487 over 1,186 months. This is a volatility forecast comparison, not a claim that volatility forecasts expected returns: volatility predicts itself, it does not predict returns, which is exactly the risk block's verdict.

8. Sensitivity analysis

Two alternatives are reported. (1) Subperiods: 1926-1989, 1990 onward, and 2012 onward (post-2011), for dividend yield, earnings yield and past volatility. (2) A less conservative overlap correction using HAC lag 0 instead of horizon - 1. The correlation estimates are unchanged by the lag choice; only inference changes.

sample	predictor	1Q	1Y	2Y	5Y
1926-1989	Dividend yield	+0.17 (+2.86)	+0.37 (+4.03)	+0.51 (+4.92)	+0.78 (+6.74)
1926-1989	Earnings yield	+0.11 (+2.17)	+0.28 (+3.29)	+0.37 (+2.95)	+0.60 (+3.17)
1926-1989	Past volatility (last	+0.14 (+2.27)	+0.02 (+0.41)	+0.01 (+0.09)	+0.01 (+0.18)
1990-latest	Dividend yield	+0.12 (+1.50)	+0.31 (+2.94)	+0.44 (+2.54)	+0.69 (+3.48)
1990-latest	Earnings yield	+0.08 (+0.83)	+0.11 (+0.72)	+0.14 (+0.64)	+0.03 (+0.11)
1990-latest	Past volatility (last	+0.07 (+0.91)	+0.14 (+1.56)	+0.17 (+1.58)	+0.14 (+0.68)
2012-latest	Dividend yield	+0.15 (+1.51)	+0.39 (+3.00)	+0.59 (+5.96)	+0.69 (+6.91)
2012-latest	Earnings yield	-0.06 (-0.54)	-0.02 (-0.06)	+0.30 (+1.41)	+0.28 (+1.32)
2012-latest	Past volatility (last	+0.18 (+1.83)	+0.21 (+1.05)	+0.24 (+2.16)	+0.00 (+0.13)

Dividend yield is strong in every era, including post-2011 (5-year 0.78 in 1926-1989, 0.69 in 1990 onward, 0.69 in 2012 onward, all significant) -- so the answer to the economic question is yes for dividend yield. Earnings yield is not: it is strong before 1990 (5-year 0.60, t = 3.17) but dead from 1990 onward (5-year 0.03, t = 0.11) and insignificant post-2011 (0.29, t = 1.33). Past volatility is mostly insignificant, with one exception (post-2011 2-year 0.24, t = 2.16). The lag sensitivity shows why the overlap correction is decisive:

predictor (HAC lag 0)	1Q	1Y	2Y	5Y
Dividend yield	+0.11 (+3.42)	+0.25 (+8.34)	+0.34 (+11.92)	+0.53 (+19.51)
Earnings yield	+0.09 (+2.95)	+0.21 (+7.10)	+0.27 (+9.26)	+0.42 (+12.01)
Past volatility (last quarte	+0.12 (+3.30)	+0.06 (+2.10)	+0.06 (+1.98)	+0.05 (+1.60)

With HAC lag 0, dividend yield's 5-year t-statistic is 19.51 instead of 3.87, and earnings yield's is 12.01 instead of 2.31. Ignoring the overlap would make nearly everything look significant; the horizon - 1 correction is what keeps the inference honest.

9. Conclusion and limitations

Answer to the economic question: predictability survives extension to 2026-10 in the full sample -- the slides' table replicates at 53 of 56 sign cells. But the two valuation ratios diverge after 1990: dividend yield stays strong in every era including post-2011, while earnings yield does not. A long-horizon investor should therefore not treat 'valuation predicts returns' as one stable fact; it is a dividend-yield fact in recent decades, and even that comes with bucket bands wide enough to make aggressive timing unwise. The signals that survive every cut are dividend yield first and cay second (significant from 1 year on in the full sample, though weaker here than in the slides). Volatility alone fails as a return predictor in every cut, while predicting realized volatility itself up to 2 years. Limitations: the underlying 'Predictor Data' files are not in this repository, so the tables above are transcribed from the notebook's executed outputs rather than re-estimated here; the slides' exact sample window and return series are inferred rather than stated, which leaves the earnings-yield gap (about 0.1 at every horizon, present under two different earnings sources) and the oil magnitude gap unresolved; cay is the supplied fixed-coefficient series, not a re-estimated cointegrating residual, so its strength depends on that choice; and post-2011 long-horizon windows are short (n = 96 for 5-year) and episode-driven.